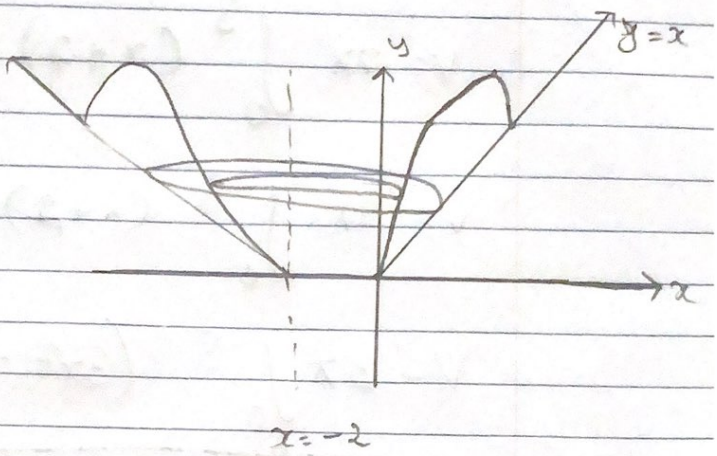
Quiz:-5

C2: $y = 4x - x^2$, $y = x$, and x-axis revolved about $x = -2$

First Quadrant



$x = 4x - x^2$

$x - 4x + x^2 = 0$

$x(x - 3) = 0$

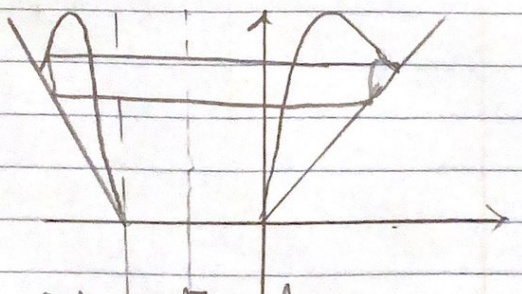
$\boxed{x = 0}$, $\boxed{x = 3}$

$$V = \int_0^3 2\pi \left[(\text{outer})^2 - (\text{inner})^2\right] dx$$

$$V = 2\pi \int_0^3 \left[(x+2)^2 - (2 + 4x - x^2)^2\right] dx$$

$$\boxed{V = 2\pi \int_0^3 \left[(x^2 + 4x + 2) - (2 + 4x - x^2)^2\right] dx}$$

C3: $y = 4x - x^2$, $y = x$ and x-axis, $x = -2$



$$V = 2\pi \int_0^3 (x+2)\left[(4x - x^2) - x\right] dx$$

$$V = 2\pi \int_0^3 (\text{radius})(\text{Top} - \text{bottom})\, dx$$

$$V = 2\pi \int_0^3 (x+2)(4x - x^2 - x)\, dx$$

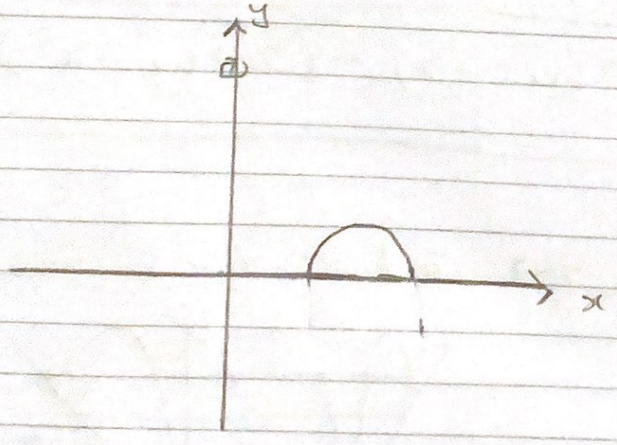
$$V = 2\pi \int_0^3 (x+2)(3x - x^2)\, dx$$

$$V = 2\pi \int_0^3 (3x^2 + 6x - x^3 - 2x^2)\, dx$$

$$\boxed{V = 2\pi \int_0^3 (x^2 - x^3 + 6x)\, dx}$$

(3)

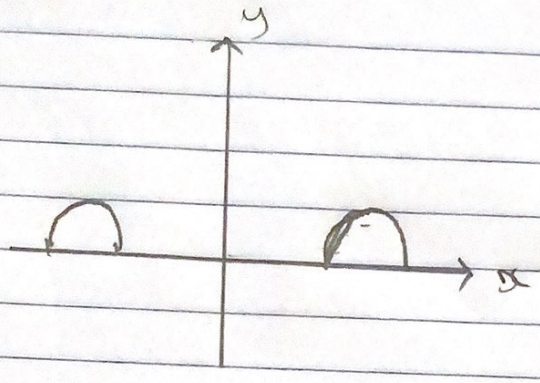
C.4: $y = 6x - x^2 - 8$, and $y = 0$, y-axis



$6x - x^2 - 8 = 0$

$x(6 - x) = 8$

$\boxed{x = 8}$ $\boxed{x = -2}$


y
x